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Toward the Establishment of a Data-Driven Learning Model: Role of Learner Factors in Corpus-Based Second Language Vocabulary Learning

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Abstract

We investigated how learner factors, such as vocabulary proficiency, strategy use, and working memory, are associated with successful corpus-based second language (L2) vocabulary learning, in which learners are encouraged to analyze and explore large, structured collections of authentic language data (i.e., corpora) to resolve their lexical issues (i.e., data-driven learning [DDL]). After measuring L2 vocabulary proficiency and working memory capacity, 35 South Korean college students performed a DDL activity during an English reading task using a think-aloud protocol to document their strategy use. Through this we identified participants' lexical inferencing strategy use, including DDL-focused strategies, based on qualitative coding. Using path analysis, we identified that participants' DDL-focused strategy use largely influenced their vocabulary acquisition and retention, highlighting the pedagogical advantages of these strategies for successful DDL. We found that participants' L2 vocabulary proficiency and working memory contributed to their vocabulary acquisition and retention, indicating the roles of these factors in managing cognitive load in DDL. Future investigation into the causal relationship between improved working memory and corpus-based L2 vocabulary learning and the role of other learner factors, including motivation and learning style, is needed to extend our understanding of DDL.

Keywords: corpus use; data-driven learning; strategy use; vocabulary learning; vocabulary proficiency; working memory

IN CORPUS-BASED SECOND LANGUAGE (L2) learning, also known as data-driven learning (DDL; Johns, 1994), learners are encouraged to analyze and explore corpora (i.e., structured collections of authentic language data) to resolve linguistic issues (Sinclair, 2004). DDL has become increasingly popular, as it offers a vast amount of comprehensible language input to learners (see Krashen, 1985, for his Input Hypothesis). Moreover, corpora, and their analysis tools, have become more available and accessible (e.g., Biber, Conrad, & Reppen, 1998; Lee, Lee, & Sert, 2015; Sinclair, 2004). This growing popularity has been supported by both theory and empirical evidence. First, corpus use has been shown to allow learners to construct their L2 knowledge independently by exploring compiled linguistic data, such as concordance lines that provide multiple examples of how a target word is used (Johns, 1994). Second, corpus tools (i.e., concordancers) display the typed item in the center of multiple concordance lines, a format called *key word in context* (KWIC; see Figure 1). This heavily exposes learners to target linguistic items (Ellis, 2002). Such exposure makes the target items more salient (Chapelle, 2003) and thus increases the possibility of learner attention to and acquisition of the target items (Schmidt, 2001). Third, cumulative empirical evidence has supported the effectiveness of DDL as an L2 learning approach. For example, based on 64 studies, Boulton and Cobb (2017) found that DDL was largely effective in L2 learning in general (d = .95) and that its effectiveness was valid across different DDL types (e.g., paper based and computer based) and different aspects of language learning (e.g., reading, writing, vocabulary, and grammar). It should be noted that in this study we focus only on paper-based DDL for lexical inferencing, with the target aspect being meaning-recall knowledge.

Despite the evidence for DDL's overall effectiveness in L2 vocabulary learning (see Lee, Warschauer, & Lee, 2019b), one cannot overlook the wide variation among learner success in general L2 learning (see Dörnyei, 2005). For example, Lee, Warschauer, and Lee (2019a), who used a data mining technique to uncover different L2 vocabulary learning patterns from experimental data, found that learners responded differently to DDL, with significant variations in their L2 vocabulary gains. As another example, a recent meta-analysis (Lee et al., 2019b) found that L2 proficiency had a statistically significant effect on corpora-based L2 vocabulary learning. Among learner factors influencing L2 vocabulary learning, it has been long assumed that cognition-related factors play significant roles in corpus-based vocabulary learning because of the heavy cognitive load involved in DDL. These factors are important because learners may need to autonomously search materials for target linguistic items while being immersed in language data, some of which may be beyond their comprehension (Boulton, 2009a; Lee et al., 2019b; Lee, Warschauer, & Lee, 2017). Thus, in addition to L2 proficiency, other learner factors—such as strategy use and working memory—may play significant roles in easing learners' cognitive loads. However, in-depth investigations on how

learners differentially construct their L2 vocabulary knowledge during DDL activities is largely absent from the literature (Boulton, 2009a; Lee et al., 2017). To address this gap, the current study investigated the role of such learner factors, focusing on L2 proficiency, strategy use, and working memory in corpus-based L2 vocabulary acquisition and retention.

RESEARCH ON LEARNER FACTORS IN CORPUS-BASED L2 VOCABULARY LEARNING

As illustrated in Figure 2, the lexical processing model proposed by de Bot, Paribakht, and Wesche (1997) points to the important role of cognition in L2 vocabulary learning. According to this model, a learner may proceed in the following steps when confronted with target vocabulary: (a) the mental lexicon determines if written input (i.e., a given word) is unknown, (b) the input is decoded, (c) the target word's string of letters (i.e., form) needs to be matched with lexemes (i.e., the sets of inflected forms of the word) in the mental lexicon, which then have to be matched with the syntactic and semantic features of the target word, and (d) finally, comprehension of the target word will be successful if the lemmas (i.e., the canonical forms of the words) are connected with one or more concepts. As de Bot et al. highlighted, the interactions between these steps may not constitute a simple linear relation, but rather a complex process that requires various types of strategies and knowledge sources to bridge the gap between the form and meaning of a target word. Considering that multiple examples are given to learners for their lexical inferencing in corpus-based vocabulary learning, it is evident that cognition-related learner factors do, indeed, matter in successful DDL.

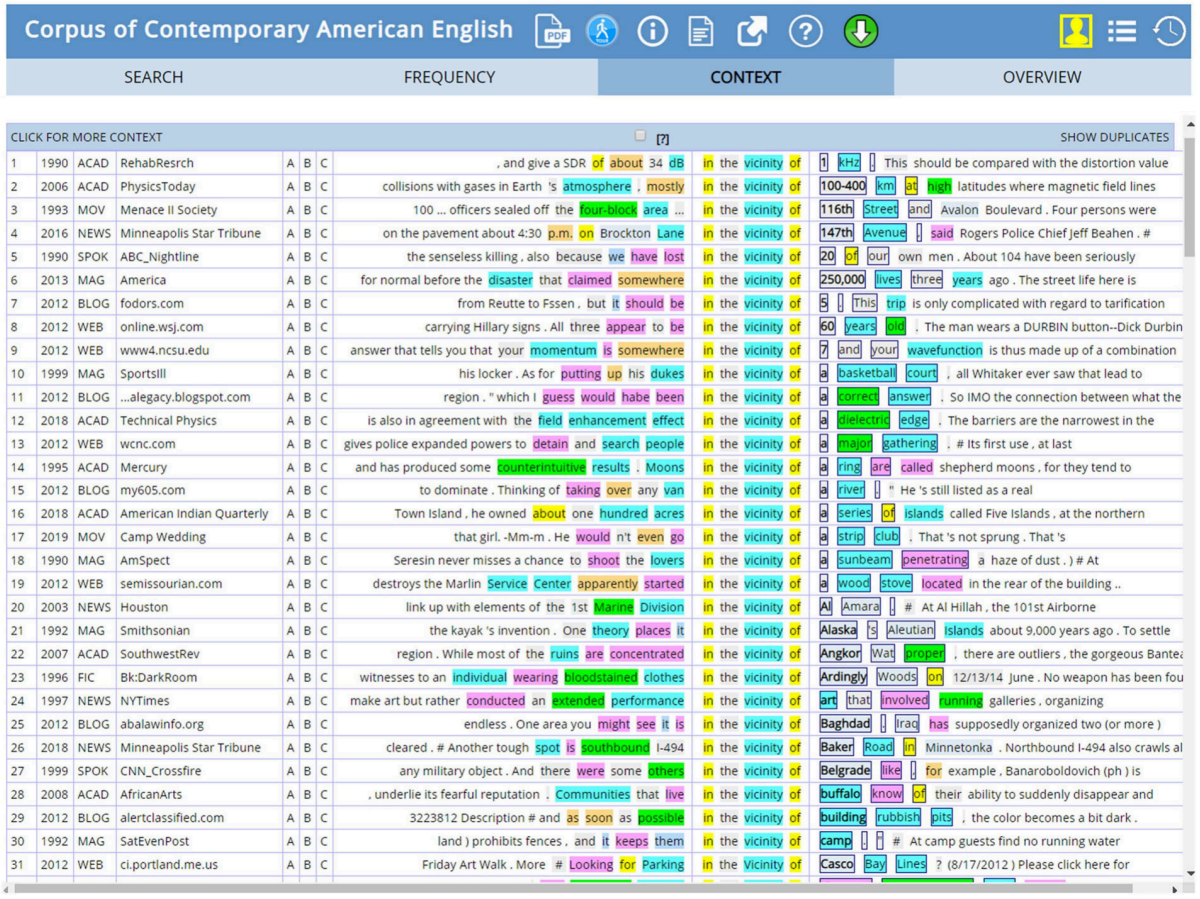


FIGURE 1

A Snapshot of Concordance Lines From the Corpus of Contemporary American English Presented by Brigham Young University (Davies, 2008)
[Color figure can be viewed at wileyonlinelibrary.com]

Note. Shown are concordance lines of *in the vicinity of*. These lines are sorted right and aligned by the target vocabulary item. In addition, the target expression is highlighted and is thus visually salient. [Color figure can be viewed at wileyonlinelibrary.com]

Moreover, as Johns (1991, 1994, 1997) and Lewis (1993) suggested, a DDL activity is not merely reading concordance lines containing target vocabulary, but a learning process that can be considered “an active, creative, and socially interactive process” (Rüschoff & Ritter, 2001, p. 223) that contains the following three stages: observe, hypothesize, and experiment. First, learners observe and research the L2 learning materials. Next, learners build a hypothesis about language features, such as contextual meaning and syntactic usage. Finally, learners test their hypothesis through practice, improvisation, or classification. The learning process of corpus-based L2 vocabulary learning involves continuous cognitive efforts to observe, hypothesize, and experiment with multiple inferences for successful vocabulary learning. For this reason, this process may be influenced by cognition-related learner factors, such as L2 proficiency, strategy use, and working memory, each of which will be discussed subsequently in terms of their relevance to corpus-based L2 vocabulary learning.

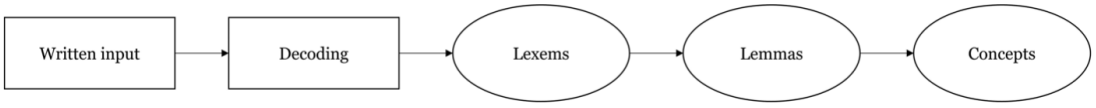


FIGURE 2
Second Language Lexical Processing Model
Note. Adapted from de Bot et al. (1997).

L2 Proficiency and Corpus-Based L2 Vocabulary Learning

In his review, Boulton (2009a) documented that some researchers had claimed that DDL placed a heavy cognitive load on learners because of its content and KWIC format (i.e., multiple concordance lines of target vocabulary). They had suggested that DDL could be only beneficial for proficient learners. For example, multiple concordance lines shown in the corpus analysis programs (see Figure 1) are randomly selected, regardless of levels of difficulty and relevance (Boulton, 2009a; Lee et al., 2019b). To allay these concerns, researchers have suggested using customized (e.g., Cobb, 1997; Lee et al., 2015) or graded reader corpora (e.g., Allan, 2009) that provide simplified (e.g., Poole, 2012) or preselected (e.g., Frankenberg–Garcia, 2014; Lee & Lee, 2015) concordance lines. Generally, these suggestions have indicated that the nature of concordance lines may be challenging, and providing more suitable language input would be helpful, particularly for low-proficiency learners.

In their meta-analysis, Lee et al. (2019b) concluded that DDL was generally effective for learners with different L2 proficiency levels, but that learners with higher L2 proficiency benefited the most from DDL for vocabulary learning. Lee et al. (2017, 2019a) offered similar conclusions, pointing to the complex role of L2 proficiency in corpus-based L2 vocabulary learning. First, Lee et al. (2017) conducted a corpus-based experiment, where L2 learners received concordance lines as glossary information, and found that participants demonstrated

higher L2 vocabulary gains on average in the treatment condition (i.e., they received concordance lines as glossary information) than in the control condition (i.e., no glossary information was received). Their achievement was significantly associated with their L2 proficiency. Second, Lee et al. (2019a) reanalyzed their previous data (from Lee et al., 2017) to investigate at the individual level and employed data mining techniques to reveal hidden patterns. The results revealed two different groups of learners based on vocabulary gains. The group of learners with higher L2 vocabulary gains was found to have significantly higher L2 proficiency than the other group. Taken together, as several other researchers have argued (Boulton, 2009a; Flowerdew, 2015; Leńko–Szymańska & Boulton, 2015), Lee et al. (2017, 2019a) found that DDL is generally effective across different proficiency levels, even for lower level (e.g., Boulton, 2009b) or beginner level (e.g., Vyatkina, 2016) learners; however, its effectiveness increases among high-proficiency learners.

Strategy Use and Corpus-Based L2 Vocabulary Learning

Johns (1991, 1994) highlighted that the DDL approach is based on inductive learning strategies, as learners observe linguistic input, perceive similarities among and differences across concordance lines, and hypothesize and test their lexical inferences (i.e., the observe, hypothesize, and experiment stages). Furthermore, Sun (2003) found that learners who were familiar with inductive learning and thinking strategies tended to explore concordance lines better. Still, there is little empirical evidence explaining how and when learners use strategies in successful DDL.

Partly because strategy use is a malleable and teachable factor (Schmitt, 2000, 2008), it is considered one of the most important learner factors in L2 vocabulary learning (see Tseng, Dörnyei, & Schmitt, 2006; Tseng & Schmitt, 2008). Moreover, there have been continuous efforts to investigate learners' use of inferencing strategies in L2 vocabulary learning (e.g., Anvari & Farvardin, 2016; Fraser, 1999; Nassaji, 2003; Shen, 2018).

Nassaji and colleagues have conducted several empirical studies to explore the role of lexical inferencing in DDL. Based on previous studies (e.g., Haastrup, 1991; Hu & Nassaji, 2012; Nassaji, 2003), Hu and Nassaji (2014) defined 12 lexical inferencing strategies that could be divided into four categories: (a) form-focused strategies, that is, analyzing, associating, and repeating, (b) meaning-focused strategies, that is, using textual clues, prior knowledge, and paraphrasing, (c) evaluating strategies, that is, making inquiries, confirming/disconfirming, and commenting, and (d) monitoring strategies, that is, stating failure or difficulty, suspending judgment, and reattempting (see Hu & Nassaji, 2014, for detailed definitions and examples of

each lexical inferencing strategy). Dividing learners into successful and unsuccessful groups based on their lexical inferencing skills, they found a statistically significant association between frequent use of monitoring strategies and successful lexical inferencing.

This set of lexical inferencing strategies has been widely adopted in studies on strategy use in L2 vocabulary learning (e.g., Anvari & Farvardin, 2016; Hermagustiana, 2018). Hermagustiana (2018) replicated Hu & Nassaji (2014), using a reading task with 10 target words. Findings from a think-aloud protocol confirmed the use of these 12 strategy types and four major strategy categories. In addition, Anvari and Farvardin (2016) found that the quality of learners' strategy use played an important role in successful lexical inferencing.

Since DDL does not only require learners to explore multiple sentence contexts, but also involves multiple lexical inferencing processes, we hypothesized that there would be unique DDL-related strategies that influence the level of success in lexical inferencing, in addition to the previously defined lexical inferencing strategies. In addition, DDL may place a heavier cognitive load on learners' lexical inferencing than vocabulary learning in a single context because of additional learning processes; therefore, management of the resulting cognitive load may be related to working memory, to which we turn next.

Working Memory and Corpus-Based L2 Vocabulary Learning

Defined as "the temporary storage and manipulation of information that is assumed to be necessary for a wide range of complex cognitive activities" (Baddeley, 2003, p. 189), working memory is one of the major factors that contribute to individual differences in L2 learning (e.g., Martin & Ellis, 2012; Williams, 2012; see Linck et al., 2014, for a review). Furthermore, working memory is one of the extensively investigated learner factors in the field of education and appears to be malleable and teachable (Tsai, Au, & Jaeggi, 2016). Recent systemic reviews (Au et al., 2015; Melby–Lervåg & Hulme, 2013) revealed that working memory trainings in general are effective in enhancing working memory capacities.

When it comes to language learning, learners' working memory capacity is often called verbal working memory (Gathercole et al., 2006). It should be noted that this construct is relatively new and underresearched in the field of L2 acquisition, and thus more research is needed on its role to further understand what determines successful L2 lexical inferencing (Hu & Nassaji, 2012). Recently, Kim (2017) found that working memory is believed to be both directly and indirectly related to vocabulary learning as part of foundational cognition. Likewise, we believe that learners' cognitive capacities are crucial for corpus-based L2 vocabulary learning, not only in terms of "memory storage, attentional control, and manipulation of information in the service of complex cognition" (Tsai et al., 2016, p. 69), but also in "encoding, maintenance, and manipulation of speech-based information" (Gathercole et al., 1992, p. 887).

PRESENT STUDY: A HYPOTHESIZED MODEL OF DATA-DRIVEN LEARNING

The present study investigates the role of learner factors in corpus-based L2 vocabulary learning using a mixed method, with the aim of establishing a model of DDL. To this end, we explore the following research questions:

RQ1. What types of lexical inferencing strategies are used, and how are they used by learners in DDL activities, as investigated through a qualitative approach (i.e., a think-aloud protocol and qualitative coding)?

RQ2. How do learner factors, such as L2 vocabulary proficiency, strategy use, and working memory, relate to successful corpus-based L2 vocabulary acquisition and retention, as investigated through a quantitative approach (i.e., a path analysis with observed variables)?

RQ3. Does our hypothesized model of DDL (see Figure 3) fit the collected data, such that L2 vocabulary proficiency and strategy use directly and indirectly contribute both to L2 vocabulary acquisition and retention, and working memory directly contributes to vocabulary acquisition while indirectly contributing to retention?

In addition, we have the following hypotheses:

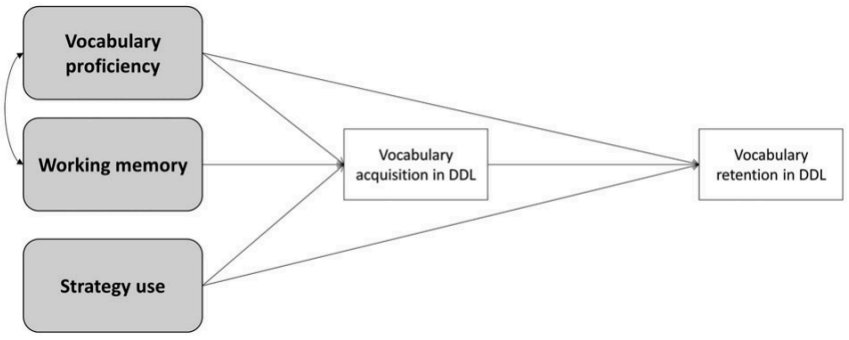


FIGURE 3
A Hypothesized Model of Data-Driven Learning (DDL)

H1. Learners will demonstrate unique DDL-focused strategy use as a task-specific cognitive learner factor (Johns, 1991, 1994; Lewis, 1993), and these strategies will be related to successful DDL (e.g., Sun, 2003).

H2. Learners with higher L2 vocabulary proficiency and working memory will benefit more from DDL than those with lower proficiency and working memory (Gathercole et al., 2006; Lee et al., 2017, 2019b).

H3. L2 vocabulary proficiency and working memory will be closely related, and they will help to manage cognitive load as general cognitive learner factors (e.g., Gathercole et al., 2006; Kormos & Sáfár, 2008).

H4. Working memory will not have a direct contribution to retention, because it is temporal in nature (Baddeley, 2003).

METHOD

Participants

A total of 35 South Korean undergraduate students with Korean as their first language (L1) participated in this study.[1] These students were all in their third year of the 4-year undergraduate study program. Except for 1 student majoring in Economics, 34 students were English education majors, with the latter generally reaching a high-intermediate level of English proficiency by the end of their undergraduate study. Their ages ranged between 19 (sophomores) and 21 (juniors and seniors), and they generally had around 10 years of formal English-learning experience. In an elective English teaching-related course with 60 students from a wide range of majors, a brief introduction of the research, including its purpose and objective, procedures, and compensation, was announced in the first class, and the participants volunteered after completing the informed consent process. One participant had to withdraw from the study due to personal reasons, so 34 students, consisting of 9 male and 25 female students, completed all necessary materials and tasks and were compensated for their time.

Reading Passage, Target Vocabulary, and Concordance Lines

To ensure successful DDL, learners should be able to comprehend the given reading passage to infer and acquire the meanings of target vocabulary. Thus, we first chose a passage entitled "What Didn't Come to Pass" (475 words) excerpted from Cunningham, Moor, and Carr (2003). Second, we analyzed the text and selected nine target vocabulary items, including the three verbs *crack*, *traipse*, and *tuck*; adjectives *dodgy*, *lumbering*, and *mucky*; and nouns *cryogenics*, *double-glazed windows*, and *grannies*. We chose the reading passage and selected the target vocabulary items in light of the test results of previous studies (Lee et al., 2017, 2019a) whose participants had profiles similar to those of the current study. Third, we retrieved lists of concordance lines of each target vocabulary item from the Corpus of Contemporary American English (COCA; Davies, 2008) to select the five most comprehensible and suitable sample sentences for the participants.[2] The selected concordance lines were displayed in the KWIC format, and the target vocabulary items were bolded,

italicized, and underlined. The reading passage (http://hansol6461.dothome.co.kr/ddl/text.htm) and examples of selected concordance lines (http://hansol6461.dothome.co.kr/ddl/ddl.htm) can be accessed at the first author's personal website.

Measures

L2 Vocabulary Proficiency. To measure the participants' L2 vocabulary proficiency, we used the Vocabulary Size Test developed by Nation and Beglar (2007). As a paper-based multiple-choice vocabulary test, it measures L2 vocabulary knowledge by asking learners to select one of the four given choices (words, expressions, or phrases) that best match the target word. For nonnative speakers of English, the test consists of 140 items that cover 14,000-word families sampled from British National Corpus (BNC) frequency lists. The maximum possible score for the test is 140. In terms of validity, the reported Rasch-based reliability measure is .96 (Beglar, 2010).

Working Memory. A listening span task (adapted from Martin & Ellis, 2012) was used to measure the participants' working memory using the researcher's laptop equipped with E-Prime 2.0 (Psychology Software Tools, Inc., 2012). The participants listened to sets of sentences that ranged from two to four sentences each and had to decide whether each sentence was grammatically correct. After listening to each set of sentences, they were asked to recall the last word, which was monosyllabic, of each sentence. For a practice trial, they listened to three sets of two sentences along with the feedback. They heard a total of 12 sets: four sets of two, three, and four sentences in random order. Therefore, the maximum possible score for this task was 36. Cronbach's alpha (α) across these 36 items was .76.

Vocabulary Tests. Paper-based meaning-recall vocabulary tests were conducted before, during, and after the DDL task in 2-week intervals to measure participants' prior knowledge, vocabulary acquisition, and retention, respectively. In these tests, the participants were asked to give the meaning of the target vocabulary items either in Korean or English. The nine target vocabulary items were alphabetically ordered, unlike the order of the items in the reading passage. To ensure the reliability of the scoring, 10 vocabulary posttests (29% of 34 vocabulary posttests) were randomly selected and scored by two of the authors. A total of two points were allotted for each item, with two points awarded for the correct meaning of vocabulary items either in L1 or L2, one point for a partially correct answer (e.g., many wrote down *fake* for the target vocabulary item *dodgy*), and zero points for an incorrect answer. Therefore, the maximum possible score for each vocabulary test was 18. Cohen's kappa coefficient (k) for the interrater reliability was .92 (SE = .05, $p < .001$).

Reading Comprehension Test. A reading comprehension test was used to ensure that students' DDL activities did not interfere with their understanding of the text. The students were required to summarize the text they had read. To ensure high interrater reliability, two of the authors scored 10 randomly selected reading tests (29% of 34 reading comprehension tests) together using a four-point scale. Cohen's kappa coefficient (*k*) for interrater reliability was .88 (SE = .16, $p < .001$).

Procedure

During the first session, the participants completed a consent form and then received a short introduction on the study. Next, the participants completed the vocabulary size test and the pretest on the target vocabulary. Two weeks later, in the second session, the researcher and a research assistant met each student individually. First, the participants completed the listening span task to measure their working memory. Second, they completed a brief training session wherein they watched a short video clip where a researcher demonstrated a DDL activity using a think-aloud protocol. Then, the participants received a list of concordance lines and were asked to try lexical inferencing using the think-aloud protocol they just watched. The researcher provided feedback when necessary but did not explicitly teach the strategies. Once a student successfully performed the task, they began the main DDL activity. They received the reading passage and additional papers with selected concordance lines. They were asked to read the passage and try to infer the meaning of target vocabulary items by consulting the concordance lines. As they were trained, they verbalized their lexical inferencing procedures throughout the task, which was untimed. After the main DDL activity, they took the vocabulary posttest (i.e., acquisition test) and reading comprehension test. Two weeks later, in the third session, the participants took the vocabulary follow-up test (i.e., retention test).

DATA ANALYSIS AND RESULTS

To achieve the research goals, we used a mixed-method approach. The qualitative component included observing, coding, and analyzing learners' L2 lexical and DDL strategy use. The quantitative component included measuring learner factors, examining L2 vocabulary acquisition and retention, and identifying associations among factors to fit the hypothesized model.

Qualitative Component: L2 Lexical Inferencing and Data-Driven Learning Strategy Use

To investigate strategy use in corpus-based L2 vocabulary learning, we used a think-aloud protocol, which has been widely used in applied linguistics to investigate learners' thought processes while executing an L2 task (Ericsson & Simon, 1993). Thus, each participant was

asked to verbalize their thoughts during their DDL activities. Before performing the task, they watched a video clip recorded by the first author on how to perform a think-aloud protocol. During their activities, they received feedback when necessary.

TABLE 1
The 12 Lexical Inferencing Strategies in Corpus-Based L2 Vocabulary Learning

Category	Freq.	%	Strategy	Freq.	%
Form-focused strategies	158	14	Analyzing	45	28
			Associating	66	42
			Repeating	47	30
Meaning-focused strategies	247	21	Using textual clues	166	67
			Using prior knowledge	28	11
			Paraphrasing	53	22
Evaluating strategies	201	17	Making inquiry	27	13
			Confirming/disconfirming	22	11
			Commenting	152	76
DDL-focused strategies	562	48	Exploring	269	48
			Cross-checking/double-checking	173	31
			Synthesizing	120	21
Total	1,168	100			

Note. DDL = data-driven learning.

Think-aloud protocols are often combined with other methods such as video-recording to triangulate findings (Deschambault, 2018), as learners often do not verbalize all of their thought processes. In the present study, the think-aloud process was video-recorded with the students' consent. The video data helped the authors to check visual clues of the learners' DDL activities, such as eye-gaze patterns.

To ensure intercoder reliability, two of the authors qualitatively analyzed 10 randomly selected video clips (29% of 34 video clips) to determine qualitative codes and themes. We referred to the 12 lexical inferencing strategies suggested by Hu and Nassaji (2014), and used an inductive approach to identify any emerging codes. We utilized Microsoft Word's memo feature to mark and label the codes on the transcripts, and then used Microsoft Excel for the coding framework when an agreement was reached.

For the first cycle, we used both process coding (i.e., coding for a word or phrase that captures action), and simultaneous coding (i.e., providing multiple codes for the same text; Saldaña, 2016) to capture how the participants responded to the language input using strategies. In doing so, every unit of strategy use was counted once regardless of repetitions, so that we could quantify the frequencies of strategy use. We found 9 of the 12 lexical strategies, representing three categories (Hu & Nassaji, 2014) in our data: analyzing, associating, and

repeating (form-focused strategies); using textual clues, using prior knowledge, and paraphrasing (meaning-focused strategies); and making inquiries, confirming/disconfirming, and commenting (evaluating strategies). In addition, we identified three unique DDL-focused strategies: exploring, cross-checking/double-checking, and synthesizing. Table 1 displays the strategies used by the participants during the DDL activities.

For the second cycle, we used pattern coding to understand how the three newly found codes under the DDL-focused category were related to other strategies and categories. Most notably, we found that the DDL-focused strategies were used mostly between concordance lines, whereas the remaining nine strategies were used within concordance lines. Furthermore, we found that the DDL-focused category was the most frequently used, and its three strategies were used more often than the other strategies, except for strategies involving the use of textual clues and comments.

Furthermore, we found that participants used monitoring strategies, such as stating the failure or difficulty, suspending judgment, and reattempting strategies suggested by Hu and Nassaji (2014), but that they were used as DDL-focused strategies, as shown in Table 2. For example, the exploring strategy describes reading multiple example sentences to infer word meaning while judging the difficulties or relevancies of the sentences, which involves stating the failure/difficulty strategy. The cross-checking/double-checking strategy involves revisiting example sentences to check or confirm previous inferences, which includes the reattempting strategy. The synthesizing strategy describes making conclusive comments about the target vocabulary based on previously made, multiple inferences and judgments through DDL, which involves the suspending judgment strategy. After this initial phase of coding, Cohen's kappa coefficient (*k*) for the intercoder reliability reached .86 (SE = .03, $p < .001$).

TABLE 2
Three Data-Driven-Learning (DDL)-Focused Strategies

Strategy	Definition	Example
Exploring	Reading multiple example sentences to infer the word meaning while judging the difficulties or relevancies of the sentences	Target Word: *lumbering* (ID: 26-13) **[Sentence #1]** I don't get it. **[S #2]** It is an adjective and seems to relate to something old. **[S #3]** Slow? **[S #4]** Something old and slow.
Cross-checking/ double-checking	Revisiting example sentences to check or confirm previous inferences after another DDL activity	Target Word: *cryogenics* (ID: 30-13) [S #1] TW is a noun. [S #5] Hmm, TW is about freezing people. [S #1] No, this is not about that. **I need to see another sentence then**. [S #4] I think TW is about freezing and defrosting people. **Let's go back. [S #1]** It does not make sense here. **[S #3]** It is a field of research related to freezing people for medical purposes. **Oh, now it makes sense. [S #1] So, this sentence** is about a salesman who works in this field of research. Now it makes sense.

Strategy	Definition	Example
Synthesizing	Making conclusive comments about the TW based on previously made multiple inferences and judgments made by DDL	Target Word: *cracked* (ID: 6–16) Okay, in the first sentence he cannot break the level, the third sentence is about getting into the hall of fame, and the fourth sentence is about going beyond a wall or barrier. **Taken all together**, I think the meaning of TW is to go beyond or to break through a level or barrier.

Note. TW = target word. Bold text is specific to each DDL-focused strategy.

Quantitative Component

Descriptive Statistics and Correlations. After identifying and quantifying learners' strategy use, we employed path analysis to determine whether the collected data fit the hypothesized model of DDL. We first explored the relationships among the variables to understand how they work together to contribute to successful corpus-based L2 vocabulary acquisition and retention using descriptive statistics (see Table 3) and correlations (see Table 4). Considering that the sample size of the present study was not large enough to accommodate many variables, we amalgamated 12 strategy types into four groups—the form-focused, meaning-focused, evaluating, and DDL-focused strategies—to assess learners' strategy use.

The descriptive statistics are displayed in Table 3. First, the descriptive statistics indicated that, on average, the participants acquired about five or six new vocabulary items (M = 11.35, SD = 3.49) and retained about three or four new vocabulary items (M = 7.44, SD = 3.55). For L2 vocabulary proficiency, the results indicated that participants had an average vocabulary size of around 7,650-word families (M = 76.5. SD = 12.08), indicating that these learners can "deal with a range of unsimplified spoken and written texts" without difficulty (Nation & Beglar, 2007, p. 9). For working memory, the participants had an average listening span task score of 25.59 (SD = 4.79), meaning that they successfully performed about 71% of the provided 36 task items.

TABLE 3
Descriptive Statistics for Target Variables

	Mean	*SD*	Minimum	Maximum
Vocabulary posttest	11.35	3.49	2	17
Vocabulary follow-up test	7.44	3.55	2	16
Vocabulary proficiency	76.5	12.08	25	93
Strategy use				
Form focused	4.64	2.10	0	10
Meaning focused	7.26	2.55	3	14
Evaluating	5.91	3.18	0	13
DDL focused	16.53	3.46	11	25
Working memory	25.59	4.79	15	34

Note. *SD* = standard deviation; DDL = data-driven learning.

Path Analysis. Path analysis using Stata 14 (StataCorp, 2015) was employed as the primary data analysis method to determine whether the collected data fit the hypothesized model of DDL. To minimize any possible bias due to small sample, we checked several statistical assumptions for path analysis: normality of residuals, homoscedasticity of residuals, multicollinearity, model fit indices—a chi-square test, root mean square error of approximation (RMSEA), comparative fit index (CFI), Tucker–Lewis index (TLI), and standardized root mean square residual (SRMR). The model passed the necessary tests (see the Appendix for the assumption test results).

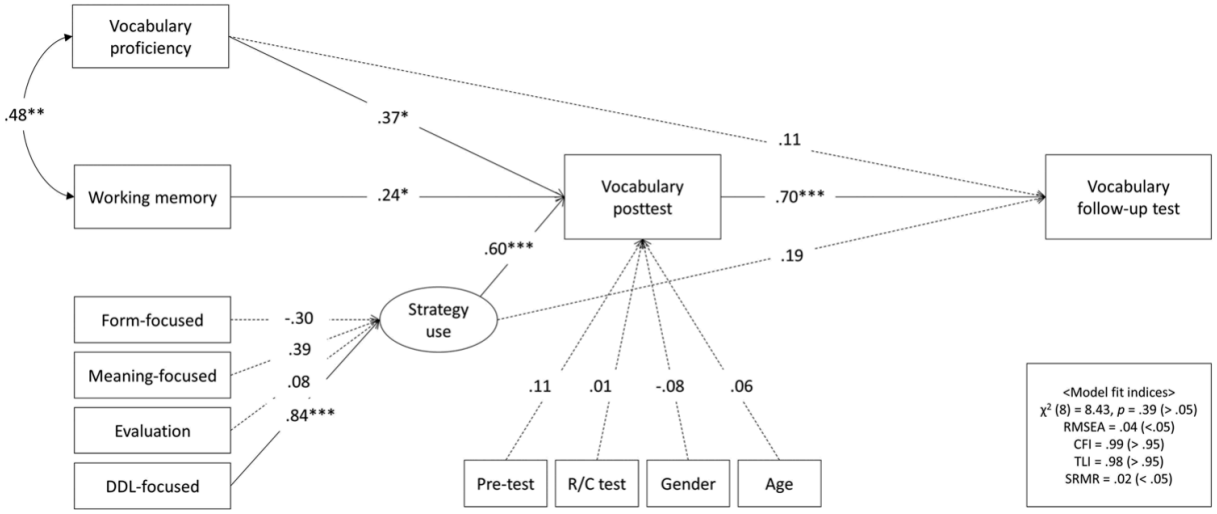


FIGURE 4

Estimated Path Analysis Model for Data-Driven Learning (DDL)

Note. R/C test = reading comprehension test. Values are standardized path coefficients for the associations of vocabulary proficiency, working memory, and strategy use to corpus-based L2 vocabulary acquisition and retention after controlling for vocabulary pretest, reading comprehension test, gender, and age. The solid lines represent statistically significant associations, whereas the dashed lines represent nonsignificant associations. The model passed the Henze–Zirkler test for multivariate normality of dependent variables at a 5% significance level. Strategy use is a composite latent variable (or a formative construct); therefore, its error variance was fixed at zero.

*p < .05, **p < .01, ***p < .001.

Figure 4 illustrates the estimated path analysis model with the associations of L2 vocabulary proficiency; working memory; and use of form-focused, meaning-focused, evaluating, and DDL-focused strategies to corpus-based L2 vocabulary acquisition and retention. In particular, the vocabulary pretest, reading comprehension test, gender, and age variables were included as control variables to remove any spurious effects. Table 5 shows the total effects (direct + indirect effects) of the independent variables on corpus-based L2 vocabulary acquisition and retention. For example, L2 vocabulary proficiency is directly associated with the vocabulary posttest, and the path coefficients shown in Figure 4 represent its direct effect. For the vocabulary follow-up test, L2 vocabulary proficiency has an insignificant direct effect, hence

the dashed line. Still, because the vocabulary posttest is directly related to the vocabulary follow-up test, L2 vocabulary proficiency may indirectly influence the vocabulary follow-up test mediated by the vocabulary posttest, and this is an indirect effect of L2 vocabulary proficiency on the vocabulary follow-up test.

TABLE 4
Correlations Between Variables of Interests

	Vocabulary posttest	Vocabulary follow-up test	Vocabulary proficiency	Strategy use				Working memory
				Form	Meaning	Evaluating	DDL	
Vocabulary posttest	–							
Vocabulary follow-up test	.84***	–						
Vocabulary proficiency	.36*	.31	–					
Strategy use								
Form focused	.16	.18	.26	–				
Meaning focused	.29	.41*	−.16	.20	–			
Evaluating	.30	.21	−.07	−.22	.19	–		
DDL focused	.46**	.44**	−.15	.24	.23	.29	–	
Working memory	.40*	.31	.48**	.35*	.11	.04	−.03	–

Note. Values are correlation coefficients. DDL = data-driven learning.
*p < .05, **p < .01, ***p < .001.

First, according to the standardized path coefficients displayed between the variables in Figure 4 and the total effects represented in Table 5, the results indicated that DDL-focused strategy use (β = .50, p < .001)—mediated by the strategy use latent variable (i.e., .84 × .60 = .50)—as well as working memory (β = .24, p < .05) and L2 vocabulary proficiency (β = .37, p < .05) were directly related to L2 vocabulary acquisition. Form-focused (β = −.30 × .60 = −.18, p > .05), meaning-focused (β = .39 × .60 = .23, p > .05), and evaluating strategies (β = .08 × .60 = .05, p > .05) were not significantly associated with vocabulary acquisition. To identify which learner factors contributed more to vocabulary acquisition, we tested the equalities of the standardized path coefficients with the Wald Chi-Squared Test. Although the effect of DDL-focused strategy use (β = .50) was descriptively larger than that of L2 vocabulary proficiency (β = .37), followed by that of working memory (β = .24), the results indicated that the differences between these three learner factors were not statistically significant (p > .05).

TABLE 5
Total Effects of Independent Variables on Corpus-Based L2 Vocabulary Acquisition and Retention

Dependent variables	Independent variables	Direct effect	Indirect effect	Total effect
Vocabulary posttest	Vocabulary proficiency	.37* (.15)	(No path)	.37* (.15)
	Strategy use			
	Form	−.18 (.13)	(No path)	−.18 (.13)

Dependent variables	Independent variables	Direct effect	Indirect effect	Total effect
Vocabulary follow-up test	Meaning	.23 (.14)	(No path)	.23 (.14)
	Evaluating	.05 (.13)	(No path)	.05 (.13)
	DDL	.50*** (.11)	(No path)	.50*** (.11)
	Working memory	.24* (.10)	(No path)	.24* (.10)
	Vocabulary proficiency	.11 (.11)	.26* (.11)	.37*** (.11)
	Strategy use			
	Form	−.06 (.05)	−.13 (.10)	−.18 (.13)
	Meaning	.08 (.10)	.16* (.08)	.24 (.16)
	Evaluating	.01 (.04)	.03 (.09)	.05 (.13)
	DDL	.16 (.13)	.35** (.11)	.51*** (.11)
	Working memory	(No path)	.17* (.08)	.17* (.08)
	Vocabulary posttest	.70*** (.12)	(No path)	.70*** (.12)

Note. Values are standardized path coefficients in the data-driven learning (DDL) model (see Figure 4) after controlling for vocabulary pretest, reading comprehension test, gender, and age. Any differences between the total effects and their parts (direct + indirect effects) are due to rounding.
*p < .05, **p < .01, ***p < .001.

Second, concerning L2 vocabulary retention, Table 5 displays the total effects of the independent variables. The vocabulary posttest (β = .70, p < .001), DDL-focused strategy use (β = .51, p < .001), L2 vocabulary proficiency (β = .37, p < .001), and working memory (β = .17, p < .05) all contributed significantly to vocabulary retention. Vocabulary acquisition (β = .70, p < .001) was directly related to retention; therefore, as a mediating variable, it enabled the independent variables to indirectly contribute to retention. In the case of meaning-focused strategy use, although it had a significant indirect effect (β = .16, p < .05), it did not have a significant total effect (β = .24, p > .05) after being combined with its nonsignificant direct effect (β = .08, p > .05). Again, we tested the equalities of the standardized path coefficients, but the results revealed that the differences between the vocabulary posttest (β = .70), DDL-focused strategy use (β = .51), and L2 vocabulary proficiency (β = .37) were not statistically significant. The total effect of working memory (β = .17) was significantly smaller than that of the vocabulary posttest (χ^2 = 21.74, p < .001) and DDL-focused strategy use (χ^2 = 4.94, p < .05). Finally, the difference between L2 vocabulary proficiency and working memory was not statistically significant (χ^2 = 1.38, p > .05).

DISCUSSION

DDL has received much attention as an effective method to improve L2 vocabulary, in that it not only provides learners with large amounts of authentic language input for linguistic inquiries (Lee et al., 2019b), but also encourages them to develop their L2 vocabulary knowledge independently (i.e., discovery learning; Flowerdew, 2015). The current study investigated the role of cognition-related learner factors, such as L2 vocabulary proficiency,

strategy use, and working memory, in determining the success of L2 vocabulary learning using DDL. Overall, we found that L2 vocabulary proficiency, DDL-focused strategy use, and working memory were significantly associated with L2 vocabulary acquisition and retention. The findings extend our understanding of the learning mechanisms behind DDL in the following important ways.

Data-Driven-Learning-Focused Strategy Use in Data-Driven Learning

First, we identified learners' use of three unique DDL-focused strategies—exploring, cross-checking/double-checking, and synthesizing—and found that they largely contributed both to L2 vocabulary acquisition (β = .50) and retention (β = .51). When compared to the other cognition-related factors—L2 vocabulary proficiency and working memory—the effect of DDL-focused strategy use was the largest for both vocabulary acquisition and retention. Moreover, its impact on vocabulary retention was statistically on a par with the vocabulary posttest (difference: χ^2 = 1.00, p > .05), which had the largest impact on retention (β = .70). These results show that DDL-focused strategy use was one of the most important factors in corpus-based L2 vocabulary learning.

The importance of DDL-focused strategy use should be emphasized because it corresponds to, and therefore empirically supports, the previously proposed learning stages of DDL: observe, hypothesize, and experiment (Johns, 1991, 1994, 1997; Lewis, 1993). For example, the benefits of learning L2 vocabulary through DDL have been attested in various L2 acquisition frameworks, and the finding of the current study showed that successful DDL learners explored and observed the concordance lines of target words (i.e., the observe stage) and made multiple inferences about target word meaning (i.e., the hypothesize stage), which led them to notice its lexical characteristics (i.e., noticing hypothesis; Schmidt, 2001). Because other concordance lines were presented, the learners actively used the additional opportunities to recheck their preliminary inferences (i.e., frequency effect; Ellis, 2002), which led them to synthesize their multiple inferences to draw conclusions (i.e., the experiment stage), leading them to become more involved in the task (i.e., involvement load hypothesis; Laufer & Hulstjin, 2001). Therefore, the current study sheds light on the ways in which learners use DDL-focused strategies, which have not received in-depth investigation to date. Further, it confirmed that these strategies substantially influence the success of corpus-based L2 vocabulary acquisition and retention.

It is thus logical to raise the question of whether DDL-focused strategies are teachable and, if so, how to teach them effectively. Concerning strategy use for L2 vocabulary learning in general, Hu and Nassaji (2014) suggested that learners should be taught to use strategies appropriate to specific contexts. We believe that this suggestion is true regarding DDL strategy

training because the KWIC format may be unfamiliar to learners (e.g., Gavioli, 2009; Lee et al., 2017). According to Lee et al. (2019b), providing training opportunities had a higher average effect size (d = .72) than providing no training opportunities (d = .58). While the effect size difference between these two categories was not statistically significant, the results do not suggest that providing training opportunities has any negative effect. Therefore, we encourage educators to implement necessary training opportunities to ensure successful DDL. Overall, in view of the characteristics of DDL and the KWIC format, it is ideal for DDL strategy training to be "clearly articulated and explicitly modelled by the teacher" (Macaro, 2001, p. 266).

Managing Cognitive Load in Data-Driven Learning

Second, we found that learners' L2 vocabulary proficiency and working memory were significantly correlated (r = .48, p < .01) and had significant total effects of similar magnitudes, both on vocabulary acquisition (β = .37 and .24, respectively; difference: χ^2 = .29, p > .05) and retention (β = .37 and .17, respectively; difference: χ^2 = .38, p > .05), confirming previous findings that learners with higher L2 proficiency and working memory benefit more from DDL than those with lower capacities (e.g., Boulton, 2009a; Flowerdew, 2015; Lee et al., 2019a; Leńko–Szymańska & Boulton, 2015, for L2 proficiency and DDL; e.g., Gathercole et al., 2006; Kim, 2017; Kormos & Sáfár, 2008; Linck et al., 2014; Martin & Ellis, 2012; Williams, 2012, for working memory with language learning in general).

We hypothesized that L2 vocabulary proficiency and working memory are general cognitive factors, and would thus have a strong influence on L2 learning in general, unlike DDL-focused strategy use, which is a task-specific and skill-based cognitive factor. For this reason, when it comes to DDL, which places more cognitive load on lexical inferencing than normal vocabulary acquisition from a single context does (e.g., Allan, 2009; Flowerdew, 2015; Lee & Lee, 2015; Lee et al., 2017), higher levels of L2 vocabulary proficiency and working memory may better manage cognitive burden during DDL performance. Previous researchers' efforts to ease the cognitive load using preselected and simplified concordance lines from customized or graded corpora are in line with the aforementioned proposition (e.g., Allan, 2009; Cobb, 1997; Frankenberg–Garcia, 2014; Lee & Lee, 2015; Lee et al., 2015; Lee et al., 2017; Poole, 2012).

Although we could not determine exactly how L2 vocabulary proficiency and working memory worked together to manage cognitive load, the findings have led us to infer that when a learner knows more L2 vocabulary word families and has a better verbal working memory, it is easier for them to explore concordance lines with different levels of difficulty and relevancy by storing inferred word meanings in their mind, to revisit concordance lines to check their inferences, and to synthesize multiple lexical inferences to draw a conclusive word meaning—

the essential stages of DDL. If this inference is correct, it may help to explain at least one of the controversial aspects regarding DDL: the role of L2 proficiency. There has been pervasive concern (see Boulton, 2009a, for a summary) that DDL may be ineffective for learners with lower L2 proficiency. However, we believe that learners' DDL-focused strategy use is likely to be the main reason why DDL is unsuccessful—not their lower L2 proficiency. Rather, it is more likely that the main role of L2 proficiency is to ease cognitive load, so that a learner can better manage the DDL task, which was highlighted in Lee et al. (2019a). This idea also aligns with Boulton and other researchers' suggestions that DDL is beneficial to all learners, but those benefits increase with higher proficiency (Boulton, 2009a, Flowerdew, 2015; Lee et al., 2019a; Leńko–Szymańska & Boulton, 2015).

Compared to L2 proficiency, working memory in L2 research is a relatively recent and underresearched topic. More importantly, whether verbal working memory is subject to improvement and whether improving working memory positively influences L2 learning have not yet been investigated thoroughly, according to Tsai et al. (2016). Nevertheless, we agree with Tsai et al.'s (2016) suggestion that working memory training will improve working memory, which in turn will positively affect general L2 learning, considering the empirically supported causal relationship between improved working memory and L1 learning (Carretti et al., 2014; Karbach, Strobach, & Schubert, 2015). Linck et al. (2014) confirmed the increasing number of investigations into the association between working memory and L2 learning; thus, as the next step, future research is required to explore the causal relationship between improved working memory and L2 learning, which will ultimately contribute to extending our knowledge of the DDL model and its cognitive components.

LIMITATIONS AND SUGGESTIONS

The present study is not without limitations. First, it was an observational study with recruited participants and no random assignment. Thus, the findings may apply to similar students in similar contexts, but generalization of its findings to a wider population may not be possible. Second, our findings would be limited to a certain form of DDL; as such, we provided the selected (i.e., filtered) concordance lines for learners' paper-based meaning-focused DDL activities. It should also be noted that the main focus of DDL tends to be more on patterns of use, so the findings of the present study, especially regarding strategy use, would differ in different formats of DDL activities—such as unfiltered online DDL activities—and this is a potential area for future research. Finally, we also suggest that researchers include and assess other learner factors in the DDL model, such as motivation and learning styles. Motivation is another dominant learner factor in L2 learning (see Tseng et al., 2006; Tseng & Schmitt, 2008), and other studies have suggested the need for further investigation of this factor to

better understand successful L2 lexical inferencing (e.g., Hu & Nassaji, 2014) and DDL (e.g., Curado-Fuentes, 2015; Lee et al., 2019a). Learning styles, which describe how a learner approaches and tackles linguistics issues (e.g., field independent, field dependent), are considered to influence learners' L2 lexical inferencing abilities (e.g., Alavi & Kaivanpanah, 2009; Flowerdew, 2008). In the present study, we found that the participants were generally motivated and field independent during their individual DDL activities, which was likely due to the compensation they received, their personal interest in the study, or the unique research setting. However, this may not be the case for other language-learning environments. Thus, taking more learner factors into consideration in a robust way may improve the model presented in this study, and thus further expand our understanding of DDL.

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APPENDIX

Testing the Assumptions of Path Analysis

Path analysis using Stata 14 (StataCorp, 2015) was employed as the primary data analysis method to examine if the collected data fit the hypothesized model of DDL. To minimize any possible bias due to small sample size, we checked several statistical assumptions for path analysis, beginning with a check for assumptions regarding sub-regression models. Shadish, Cook, and Campbell (2002) suggested that (a) the residuals (errors) be identically and independently distributed (i.e., normality of residuals; Shapiro–Wilk test [Shapiro & Wilk, 1965]; $p > .05$), (b) the variance of the residuals should be constant across all values of the independent variables (i.e., homoscedasticity of residuals; Cameron & Trivedi's test [Cameron & Trivedi 1990]; $p > .05$), and (c) the independent variables should not be linear combinations of one another (i.e., multicollinearity; variance inflation factors [VIF] for each independent variable should not be greater than five). As a robustness check, we further checked whether the joint distribution of the dependent variables is multivariate normal (Henze–Zirkler test [Henke & Zirkler, 1990] $p > .05$), which is normally suggested for a SEM model (Kline, 2012). Results indicated that the two dependent variables in our path analysis passed the Henze–Zirkler test for multivariate normality of dependent variables at a 5% significance level. Furthermore, following Acock (2013) and Kline (2012), we evaluated the model fits by a chi-square test in which the model fit should not be significantly poorer than a saturated model ($p > .05$), root mean square error of approximation (RMSEA < .05), comparative fit index (CFI > .95), Tucker–Lewis index (TLI > .95), and standardized root mean square residual (SRMR < .05).

Our hypothesized path analysis model had two dependent variables (i.e., vocabulary posttest and follow-up test). Therefore, assumption checks were conducted for each regression model. For the first model with the vocabulary posttest as the dependent variable, the data passed the Shapiro–Wilk test (normality of residuals; $p = .23$) and Cameron and Trivedi's test (homoscedasticity; $p = .58$). In addition, the mean VIF was found to be 1.70, and the VIF for each variable was less than 5.0 (Min−Max: 1.32−2.53). For the second model with the vocabulary follow-up test as the dependent variable, the data passed the Shapiro–Wilk test (normality of residuals; $p = .42$) and Cameron and Trivedi's test (homoscedasticity; $p = .33$). Moreover, the mean VIF was found to be 1.87, and the VIF for each variable was less than 5.0

(Min−Max: 1.33−2.55). Last, we found that the estimated path analysis model had acceptable model fit indices: χ^2 (8) = 8.43, p > .05; RMSEA < .05; CFI > .95; TLI > .95; SRMR < .05 (see Figure 4).

SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

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1. The sample size may not be large enough for path analysis or structural equation modeling (SEM) based on the rule of thumb for the minimum sample for multivariate analyses, such as the 10 cases per variable rule or the 5 × 2k rule (k = number of variables; see Dolnicar, 2002, for a review). However, recent simulation studies (Sideridis et al., 2014; Wolf et al., 2013) have emphasized the limitations of commonly cited rules of thumb and recommended that small sample sizes are sufficient (e.g., 30 cases for a one-latent-variable SEM model with four variables; Wolf et al., 2013). Furthermore, the required sample size for path analysis or SEM is largely affected by how much the data set satisfies statistical assumptions (e.g., multivariate normality); if the data have any missing values (Schreiber et al., 2006); and, most importantly, whether the data obtain an overall good model fit.↩
2. We used the process of selecting example concordance lines employed in Lee et al. (2017), which is as follows: (a) sample sentences should be comprehensible for students and should not have unfamiliar words and phrases, (b) sample sentences should have obvious clues to infer the contextual meaning of target vocabulary used in the passage, and (c) sample sentences that may induce faulty or irrelevant meaning inferences should be excluded.↩